

Product Requirements Document (PRD) v3.0

Daily Medical Nutrition & Fitness Tracker — iOS Native (Swift UI / SwiftData)

VERSION: 3.0
(Updated)

AUTHOR: iOS Product
Engineering

PLATFORM: iOS (SwiftUI + SwiftData +
HealthKit)

STATUS: Ready for Sprint
Execution

1. Executive Summary & Core Objective

This document outlines the production specifications for a native iOS application designed to replace paper-based daily medical nutrition and wellness tracking sheets. By moving to an iPhone-native workflow built with **SwiftUI** and **SwiftData**, logging time is reduced from 5+ minutes to **under 30 seconds per day**.

In PRD v3.0, we have introduced the **Feelings & Cravings Real-Time Tracker**—a streamlined timestamped logging module allowing users to record instant moments of hunger, cravings, fatigue, or mood with 1 tap.

2. Architecture & Scope Highlights

Confirmed Functional Upgrades (v3.0 Scope):

- **Feelings & Cravings Log (New v3.0):** Quick-tap timestamped log for hunger spikes, cravings, energy dips, and emotional check-ins (e.g., "Craving sweets at 3:15 PM").
- **Doctor/Coach PDF & CSV Export:** 1-tap generation of styled multi-page PDF reports (including feelings timeline) and CSV data exports for medical review.
- **Apple Health (HealthKit) Integration:** Bidirectional sync for hydration volume (fluid oz) and activity session durations.
- **Smart Push Notifications:** Configurable local reminders for water intake intervals and evening check-ins.
- **Multi-Activity Exercise Engine:** Full support for logging multiple independent workout sessions per day.

3. Detailed Feature Specifications

3.1 Header, Macros & Daily Controls P0 - CRITICAL

- **Date Picker:** Defaults to current day with swift arrow navigation across historical entries.
- **Live Calorie Ring:** Real-time sum of protein calories against daily goal (e.g., 520 / 500 kcal) with color states (Green = on target, Amber = over target).
- **Ketosis & Plan Compliance Toggles:** 1-tap binary switches ('Ketosis: Y/N', 'Followed Plan: Y/N').
- **Notes Field:** Multi-line plain text storage for daily health observations.

3.2 Feelings & Cravings Real-Time Tracker P0 - CRITICAL NEW V3.0

- **1-Tap Quick-Log Buttons:** Quick categories (Hungry, Craving Sweets, Craving Salty, Low Energy, Anxious/Stressed, Full/Satisfied).
- **Auto-Timestamping:** Automatically captures exact time (e.g., "Craving Sweets at 3:42 PM") with optional custom note box.
- **Daily Timeline View:** Compact chronological list displaying all feelings/cravings logged throughout the day.

3.3 Hydration Engine & HealthKit Sync P0 - CRITICAL HEALTHKIT

- **Visual Counter:** Interactive row of 8oz glass icons (64oz baseline target). Tapping glass N automatically fills 1 through N.
- **HealthKit Integration:** Writing water logs automatically creates HKQuantityTypeIdentifierDietaryWater samples in Apple Health.

3.4 Protein Log & Hunger Scale Engine P0 - CRITICAL

- **Preset Library:** Pre-seeded with items from physical sheet (*Canadian bacon, Egg whites, Broiled salmon, Turkey bacon, Chocolate Premium Protein*).
- **"Add Custom Preset" Workflow:** Entering a custom food item displays a toggle to "Save to My Presets" for future 1-tap reuse.
- **Hunger Scale Sliders:** Dual 1–10 rating controls for **Before Meal** and **After Meal** levels.

3.5 Fast Checklists & Multi-Activity Workouts P1 - HIGH

- **Fats/Veggies Quick-Tap:** Pre-filled items matching sheet + custom additions.
- **Supplements Grid:** 1-tap pill checkboxes for daily medication/supplements.
- **Multi-Activity Workouts:** Add independent workout entries with durations, synced with Apple Health.

3.6 Medical PDF & CSV Export Engine P1 - HIGH

- **Medical PDF Generator:** Formats weekly or monthly logs (including Feelings & Cravings Timeline) into a clean PDF printout for clinical review.
- **CSV Export:** Raw data export via standard iOS Share Sheet.

4. Production SwiftData Data Schema

```
import Foundation
import SwiftData

@Model
final class DailyLog {
    var id: UUID
    var date: Date
    var proteinGoal: Int
    var ketosis: Bool
    var followedPlan: Bool
    var notes: String
    var waterOz: Int

    @Relationship(deleteRule: .cascade) var proteinEntries: [ProteinEntry]
    @Relationship(deleteRule: .cascade) var workoutEntries: [WorkoutEntry]
    @Relationship(deleteRule: .cascade) var feelingEntries: [FeelingEntry] // New in v3.0

    var checkedFatsAndVeggies: [String]
    var checkedMiscItems: [String]
    var completedSupplements: [String]

    init(date: Date = Date(), proteinGoal: Int = 500) {
        self.id = UUID()
        self.date = date
        self.proteinGoal = proteinGoal
        self.ketosis = true
        self.followedPlan = true
        self.notes = ""
        self.waterOz = 0
        self.proteinEntries = []
        self.workoutEntries = []
        self.feelingEntries = []
        self.checkedFatsAndVeggies = []
        self.checkedMiscItems = []
        self.completedSupplements = []
    }
}

@Model
final class FeelingEntry { // New in v3.0
    var id: UUID
    var type: String // e.g. "Hungry", "Craving Sweets", "Low Energy"
    var timeLogged: Date
    var note: String
}
```

```

    init(type: String, note: String = "", timeLogged: Date = Date()) {
        self.id = UUID()
        self.type = type
        self.note = note
        self.timeLogged = timeLogged
    }
}

@Model
final class ProteinEntry {
    var id: UUID
    var name: String
    var time: Date
    var servingSize: String
    var calories: Int
    var hungerBefore: Int // Scale 1-10
    var hungerAfter: Int // Scale 1-10
}

@Model
final class WorkoutEntry {
    var id: UUID
    var activityName: String
    var durationMinutes: Int
    var timeLogged: Date
}

```

5. Updated Data Mapping Reference

FEATURE / USER INPUT	SAMPLE TRIGGER	DIGITAL SWIFT MODEL EXECUTION
Feelings & Cravings (New)	Tap "Craving Sweets" at 3:15 PM	FeelingEntry(type: "Craving Sweets", timeLogged: 15:15)
Feelings Note (New)	"Hungry, felt lightheaded" at 11:10 AM	FeelingEntry(type: "Hungry", note: "felt lightheaded", timeLogged: 11:10)
Header Status	Nov 13 Goal: 500 Ketosis: Y	DailyLog(date: 2026-11-13, goal: 500, ketosis: true)
Protein Meal Log	Canadian bacon 8:30 35 kcal	ProteinEntry(name: "Canadian bacon", time: 08:30, cal: 35)
Multi-Workouts	Interactive Ex. (30m) + Brisk walk (45m)	[WorkoutEntry("Interactive Ex", 30), WorkoutEntry("Brisk walk", 45)]